

PHASE 2 · PROGRESS PRESENTATION



Fatigue Awareness and Driver Engagement Routing

DOMAIN	TRANSPORTATION AND LOGISTICS
PROBLEM STATEMENT NO.	OMNI_TRANSPORT_9
PROBLEM STATEMENT	REDUCING FATIGUE-RELATED TRUCKING ACCIDENTS
TEAM NAME	bi.Variates
TEAM MEMBERS	ABHINAV BUTE, ATHARVA KALE

PROBLEM STATEMENT

Reducing Fatigue-Related Trucking Accidents

20–30%

of road accidents worldwide are estimated to involve driver fatigue or drowsiness (road-safety research incl. RoSPA)

54%+

of crashes on India's Agra–Lucknow Expressway were linked to drivers falling asleep at the wheel (UPEIDA data)

Why long, monotonous regional routes are the highest-risk environment:

1

Highway Hypnosis

Featureless, low-stimulus stretches induce a trance-like state — eyes stay open, but attention quietly disengages from the road.

2

CO2 Build-Up

In sealed cabins with recirculated air, rising in-cabin CO2 over a long haul measurably dulls alertness and slows reaction time.

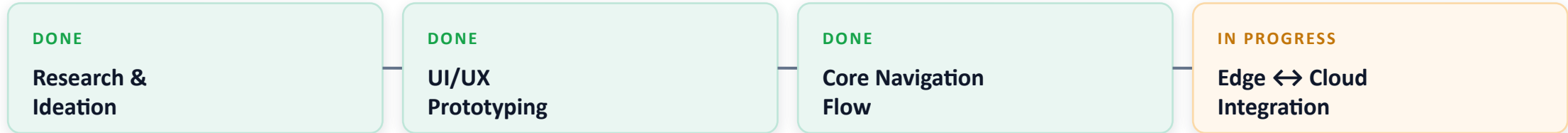
3

Microsleeps

Involuntary 1–30 second lapses where the brain briefly disengages — often unnoticed by the driver until it is too late.

Regional highway hauls combine all three factors at once — making multi-hour truck and bus routes the single highest-risk driving context.

From Concept to a Working Prototype



Built & Functional

- Complete Flutter UI across 4 core screens — Splash, Pre-Trip Check, Active Journey (Route + Fatigue Monitor)
- Working end-to-end navigation flow: onboarding → trip setup → live journey dashboard
- Route, ETA and weather data cards wired into the live-journey layout
- 3-Strike buzzer logic and risk-based brightness slider implemented client-side
- System architecture finalized — Edge/Cloud split and WebSocket contract drafted

Currently In Progress

- Google MediaPipe Face Mesh integration for real-time Eye Aspect Ratio (EAR) tracking
- FastAPI backend scaffolding for mapping, weather and LLM microservices
- Mapbox + Tomorrow.io API wiring for live routing and weather telemetry
- Conversational-AI trivia engine and voice-engagement check module

What FADER Does Behind the Wheel

1

3-Strike “Dead Man’s Switch”

Escalating buzzer alarm arms after 3 consecutive low-EAR / closed-eye signals, forcing driver acknowledgment or flagging an immediate swap.

2

Predictive CO2 Cabin-Flush

Forecasts in-cabin CO2 build-up across the remaining trip and prompts a fresh-air flush break before drowsiness sets in.

3

Location-Based LLM Trivia

The conversational engine generates route-aware trivia and check-in prompts tied to upcoming towns and landmarks to keep the driver verbally engaged.

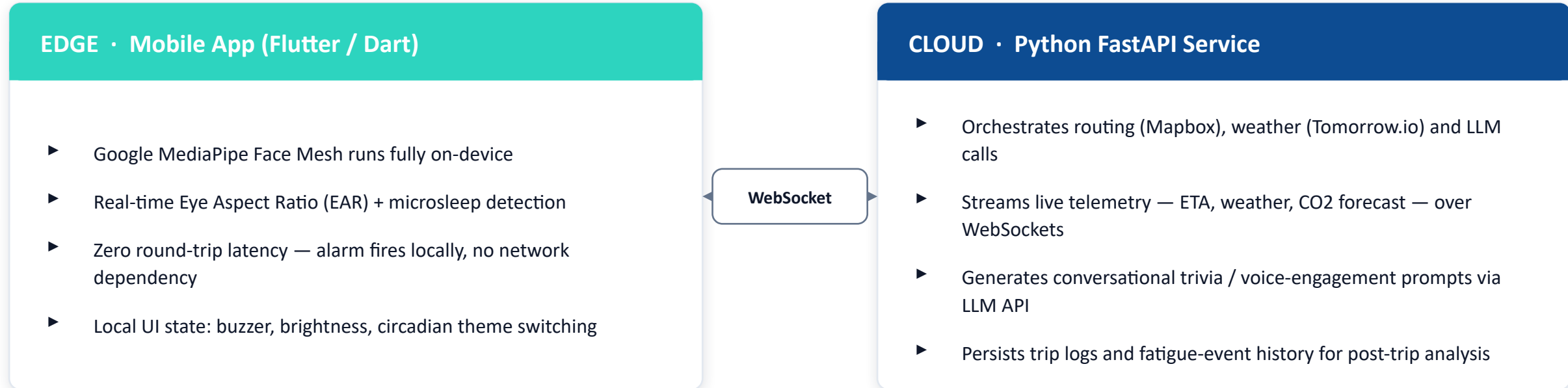
4

Circadian Light Enforcement

Auto-switches to a low-blue-light theme after sunset, and raises screen brightness inside detected high-fatigue zones to re-stimulate alertness.

Every feature intervenes before fatigue becomes dangerous — not after.

Edge-to-Cloud Split for Zero-Latency Safety



TECHNOLOGY STACK



Engineering Around Real-World Constraints

1

On-Device Inference Cost

Continuous Face Mesh inference drains battery and CPU on mid-range phones. Mitigated by throttling frame-sampling rate and downscaling the camera feed.

2

Connectivity Gaps on Highways

Weather, routing and LLM calls need cloud reachability that regional highways don't guarantee. Mitigated with WebSocket auto-reconnect and cached last-known telemetry.

3

Cross-Driver EAR Calibration

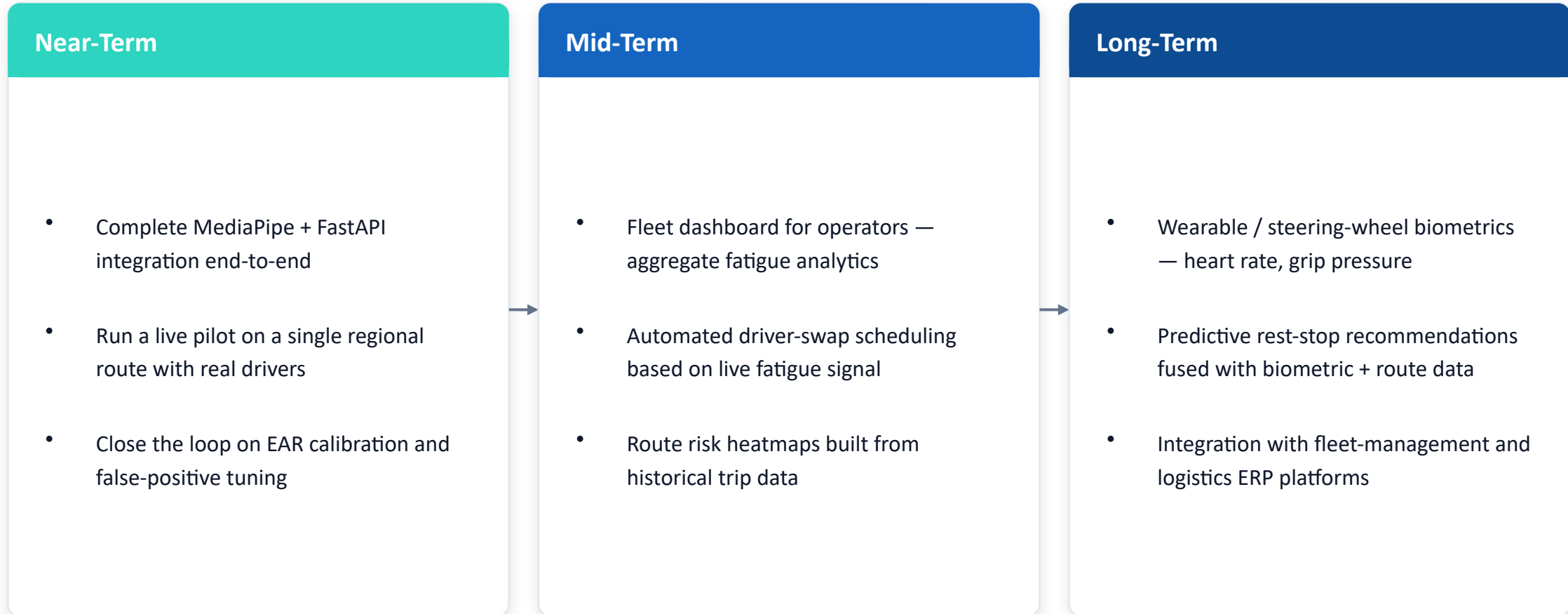
Eye Aspect Ratio thresholds shift with eye shape, glasses and lighting. Addressed with a short per-driver calibration step during Pre-Trip Check.

4

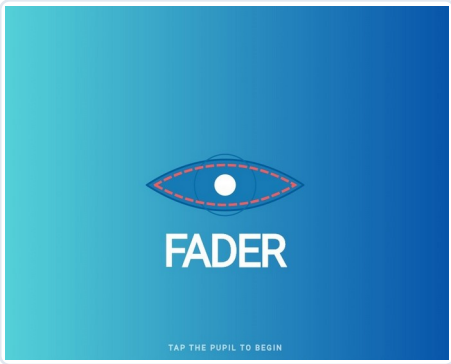
False-Positive Fatigue Signals

Head turns, glare and low light can mimic closed-eye readings. Being addressed by fusing EAR with blink duration and head-pose checks, not a single signal.

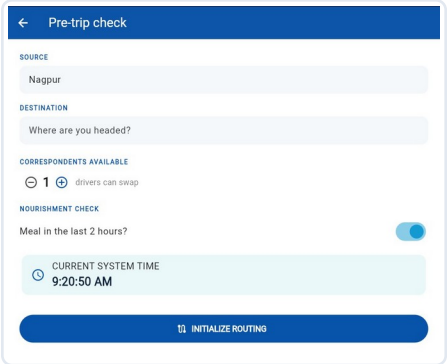
From Prototype to Fleet-Ready Co-Pilot



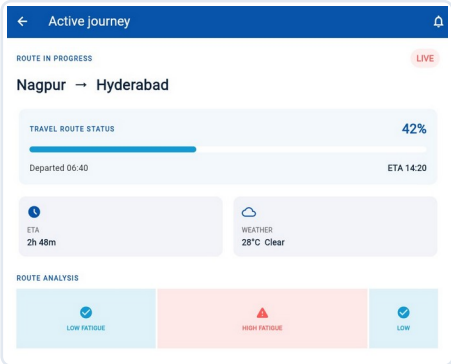
FADER in Action



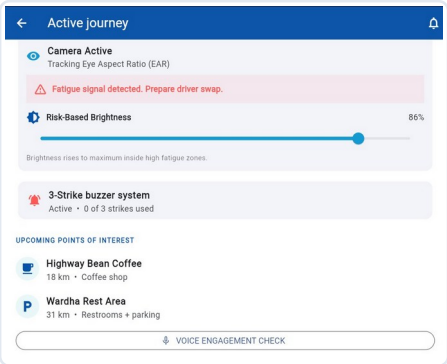
Splash Screen — Brand Entry Point



Pre-Trip Check — Route & Driver Readiness



Active Journey — Live Route & Weather Telemetry



Active Journey — Real-Time Fatigue Detection